01_EDA

May 12, 2026

1 UPDATE 1: EDA

[55]:
```python
import pandas as pd
import numpy as np

# don't display warnings
import warnings
warnings.filterwarnings('ignore')
```

[56]:
```python
df = pd.read_csv('../data/full.csv')
df.head()
```

[56]:
```
                                            Headline     Stance  \
0  Police find mass graves with at least '15 bodi…  unrelated
1  Hundreds of Palestinians flee floods in Gaza a…      agree
2  Christian Bale passes on role of Steve Jobs, a…  unrelated
3  HBO and Apple in Talks for $15/Month Apple TV …  unrelated
4  Spider burrowed through tourist's stomach and …   disagree

                                         articleBody
0  Danny Boyle is directing the untitled film\r\n…
1  Hundreds of Palestinians were evacuated from t…
2  30-year-old Moscow resident was hospitalized w…
3  (Reuters) - A Canadian soldier was shot at the…
4  Fear not arachnophobes, the story of Bunbury's…
```

1.0.1 Separating the data into related and unrelated ones

[57]:
```python
related_df = df[df['Stance'].isin(['agree', 'disagree', 'discuss'])]
unrelated_df = df[df['Stance'] == 'unrelated']
```

[58]:
```python
df['Stance'].value_counts()
```

[58]:
```
Stance
unrelated    54894
discuss      13373
agree         5581
disagree      1537
```

```
Name: count, dtype: int64
```

[59]:
```
related_df['Stance'].value_counts()
```

[59]:
```
Stance
discuss     13373
agree        5581
disagree     1537
Name: count, dtype: int64
```

[60]:
```
unrelated_df['Stance'].value_counts()
```

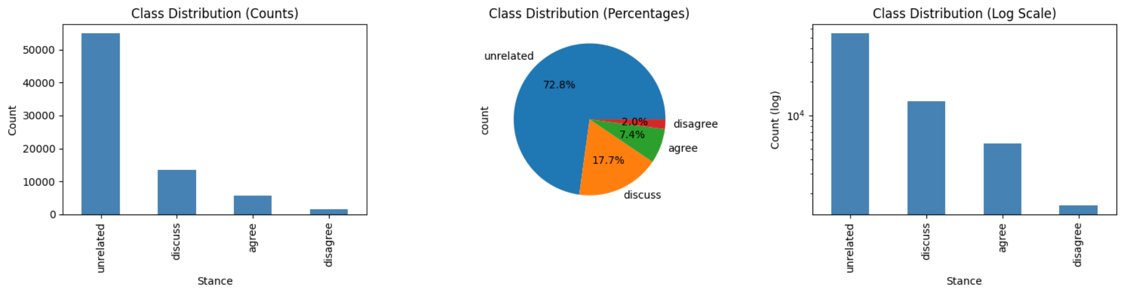
[60]:
```
Stance
unrelated    54894
Name: count, dtype: int64
```

[61]:
```
related_df.to_csv('../data/related.csv', index=False)
unrelated_df.to_csv('../data/unrelated.csv', index=False)
```

1.1 Stance distribution and Analysis

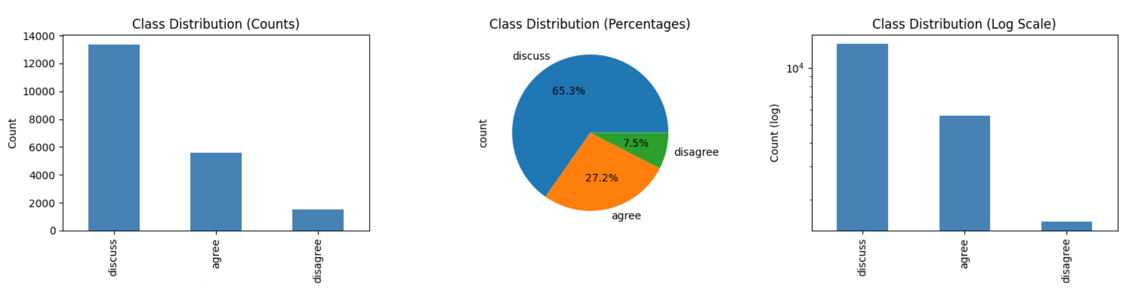
1.1.1 Entire dataset

[62]:
```
import matplotlib.pyplot as plt
import seaborn as sns

# Overall distribution
fig, axes = plt.subplots(1, 3, figsize=(15, 4))

# Bar chart (absolute counts)
df['Stance'].value_counts().plot(kind='bar', ax=axes[0], color='steelblue')
axes[0].set_title('Class Distribution (Counts)')
axes[0].set_ylabel('Count')

# Pie chart (percentages)
df['Stance'].value_counts().plot(kind='pie', ax=axes[1], autopct='%1.1f%%')
axes[1].set_title('Class Distribution (Percentages)')

# Log scale (to see minority classes)
df['Stance'].value_counts().plot(kind='bar', ax=axes[2], color='steelblue')
axes[2].set_yscale('log')
axes[2].set_title('Class Distribution (Log Scale)')
axes[2].set_ylabel('Count (log)')

plt.tight_layout()
plt.show()
```



1.1.2 Only Related

[63]:
```python
# Overall distribution
fig, axes = plt.subplots(1, 3, figsize=(15, 4))

# Bar chart (absolute counts)
related_df['Stance'].value_counts().plot(kind='bar', ax=axes[0],␣
    ↪color='steelblue')
axes[0].set_title('Class Distribution (Counts)')
axes[0].set_ylabel('Count')

# Pie chart (percentages)
related_df['Stance'].value_counts().plot(kind='pie', ax=axes[1], autopct='%1.
    ↪1f%%')
axes[1].set_title('Class Distribution (Percentages)')

# Log scale (to see minority classes)
related_df['Stance'].value_counts().plot(kind='bar', ax=axes[2],␣
    ↪color='steelblue')
axes[2].set_yscale('log')
axes[2].set_title('Class Distribution (Log Scale)')
axes[2].set_ylabel('Count (log)')

plt.tight_layout()
plt.show()
```



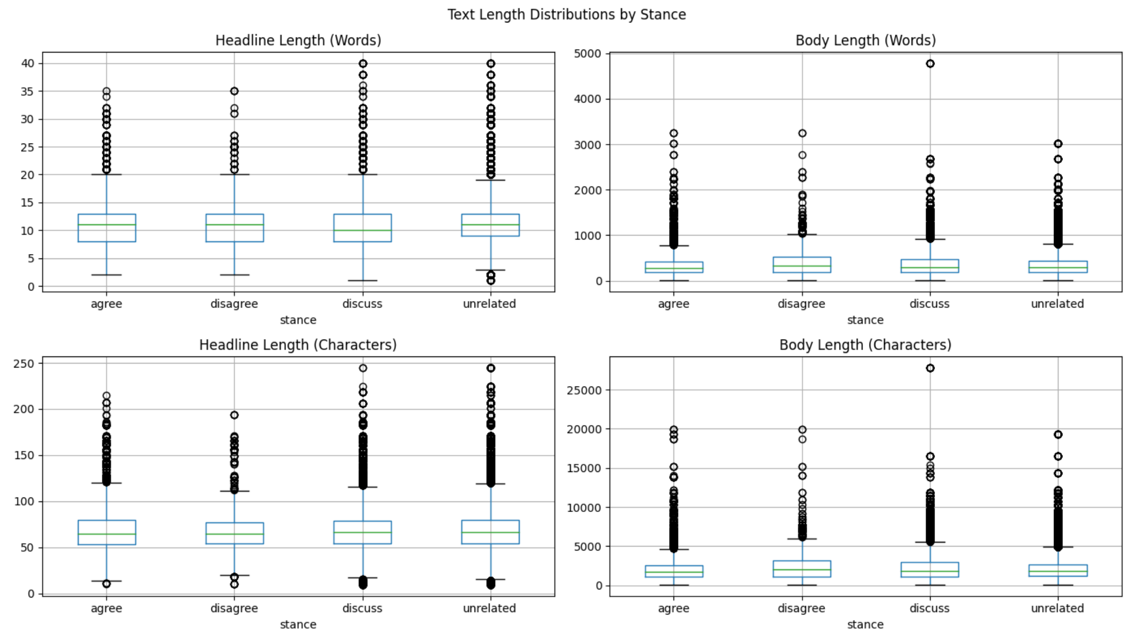
From the above distribution, we can see that the dataset is highly imbalanced, with the majority class being "unrelated".

This imbalance can pose challenges for machine learning models, as they may be biased towards the majority class. It will be important to consider techniques for handling this imbalance, such as resampling or using appropriate evaluation metrics.

1.1.3 Stance text length distribution

[64]:
```
df = df.rename(columns={'Headline': 'headline', 'articleBody': 'body', 'Stance':
 ↪ 'stance'})
```

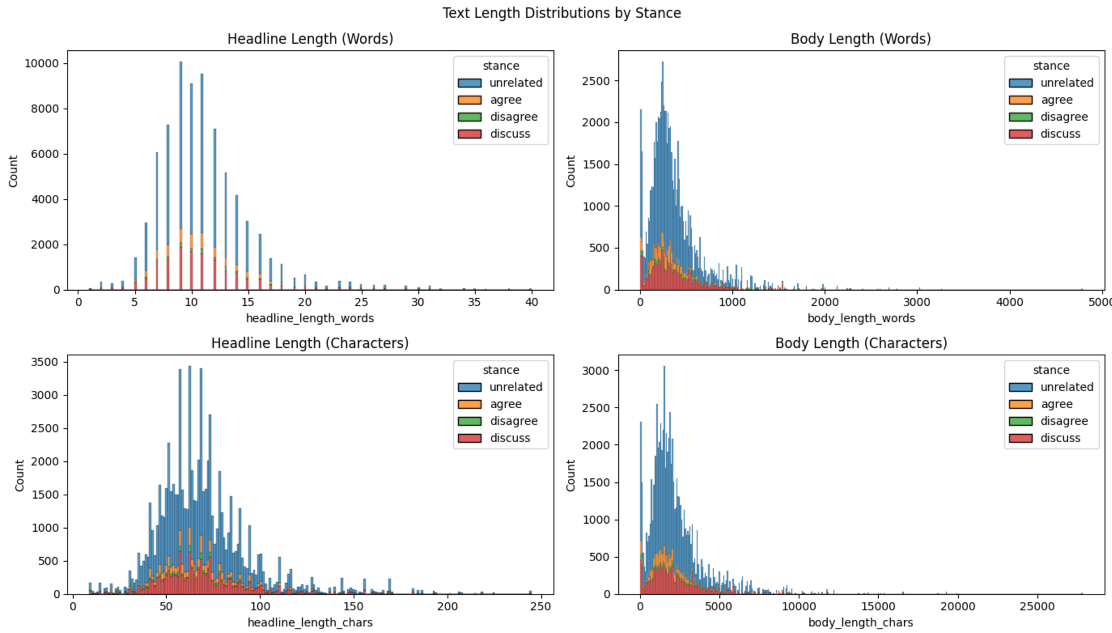
[65]:
```
# Add length columns
df['headline_length_words'] = df['headline'].str.split().str.len()
df['headline_length_chars'] = df['headline'].str.len()
df['body_length_words'] = df['body'].str.split().str.len()
df['body_length_chars'] = df['body'].str.len()

# Statistics by stance
df.groupby('stance')[['headline_length_words', 'headline_length_chars',
                      'body_length_words', 'body_length_chars']].agg(['mean',␣
 ↪'median', 'min', 'max', 'std'])

# Visualizations
fig, axes = plt.subplots(2, 2, figsize=(14, 8))

df.boxplot(column='headline_length_words', by='stance', ax=axes[0, 0])
axes[0, 0].set_title('Headline Length (Words)')

df.boxplot(column='body_length_words', by='stance', ax=axes[0, 1])
axes[0, 1].set_title('Body Length (Words)')

df.boxplot(column='headline_length_chars', by='stance', ax=axes[1, 0])
axes[1, 0].set_title('Headline Length (Characters)')

df.boxplot(column='body_length_chars', by='stance', ax=axes[1, 1])
axes[1, 1].set_title('Body Length (Characters)')

plt.suptitle('Text Length Distributions by Stance')
plt.tight_layout()
plt.show()
```



On avergage, I don't really see that much of a difference in the text length distribution across the different stances. However, there are some outliers in the "disagree" and "discuss" categories that have longer text lengths. As well as the "unrelated" category has some outliers with shorter text lengths.

[66]:
```python
# histograms

fig, axes = plt.subplots(2, 2, figsize=(14, 8))

sns.histplot(data=df, x='headline_length_words', hue='stance', ax=axes[0, 0],
    multiple='stack')
axes[0, 0].set_title('Headline Length (Words)')

sns.histplot(data=df, x='body_length_words', hue='stance', ax=axes[0, 1],
    multiple='stack')
axes[0, 1].set_title('Body Length (Words)')

sns.histplot(data=df, x='headline_length_chars', hue='stance', ax=axes[1, 0],
    multiple='stack')
axes[1, 0].set_title('Headline Length (Characters)')

sns.histplot(data=df, x='body_length_chars', hue='stance', ax=axes[1, 1],
    multiple='stack')
axes[1, 1].set_title('Body Length (Characters)')

plt.suptitle('Text Length Distributions by Stance')
```

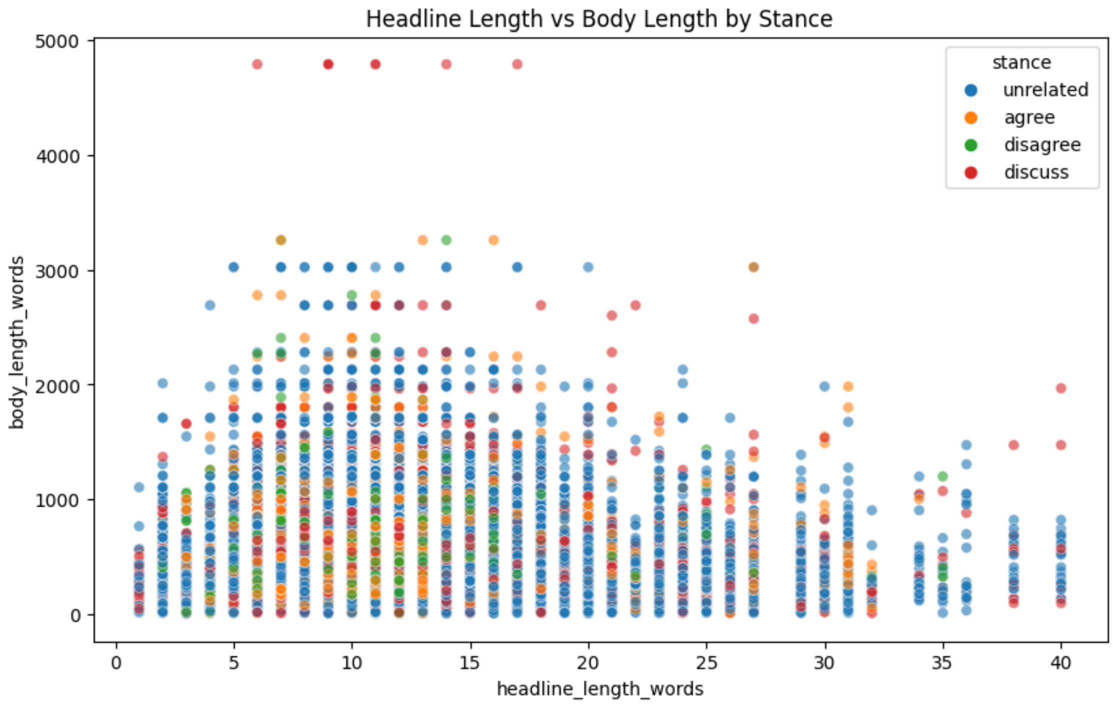
```
plt.tight_layout()
plt.show()
```



This was as expected - right skewed with a long tail as some are very long, and most falling in the shorter range.

But the interesting part is in body lengths, we see a huge spike at lower side (which is different than the spike at mean). This means that there are a lot of bodies that are very short, which is interesting. We will need to look into these short bodies and see if they are valid or if they are noise in the data.

[67]:
```
# Scatter plot of headline length vs body length colored by stance
plt.figure(figsize=(10, 6))
sns.scatterplot(data=df, x='headline_length_words', y='body_length_words',␣
 ↪hue='stance', alpha=0.6)
plt.title('Headline Length vs Body Length by Stance')
plt.show()
```

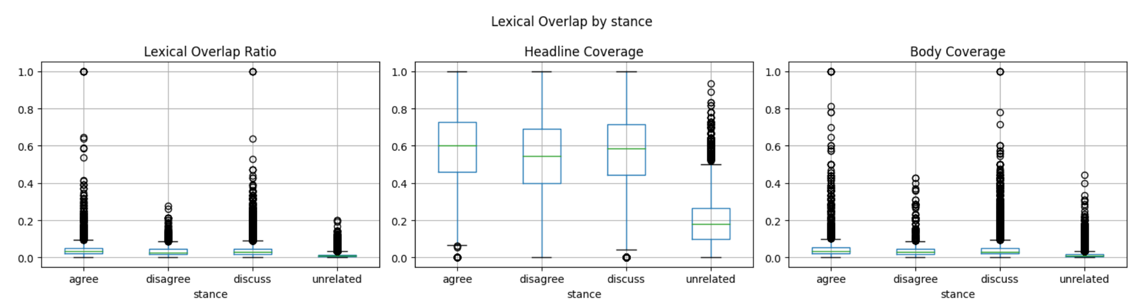


All those extra long bodies are all discusses, which is interesting. We will need to look into these long bodies and see if they are valid or if they are noise in the data.

1.1.4 Lexical Overlap

[68]:
```
# Function to calculate overlap
def calculate_overlap(headline, body):
    headline_words = set(str(headline).lower().split())
    body_words = set(str(body).lower().split())

    overlap = headline_words.intersection(body_words)
    union = headline_words.union(body_words)

    if len(union) == 0:
        return 0

    overlap_ratio = len(overlap) / len(union)
    headline_coverage = len(overlap) / len(headline_words) if␣
↪len(headline_words) > 0 else 0
    body_coverage = len(overlap) / len(body_words) if len(body_words) > 0 else 0

    return overlap_ratio, headline_coverage, body_coverage

# Apply function
```

```
df[['overlap_ratio', 'headline_coverage', 'body_coverage']] = df.apply(
    lambda row: pd.Series(calculate_overlap(row['headline'], row['body'])),
    axis=1
)

# Statistics by stance
df.groupby('stance')[['overlap_ratio', 'headline_coverage', 'body_coverage']].
 ↪agg(['mean', 'median', 'min', 'max', 'std'])

# Visualizations
fig, axes = plt.subplots(1, 3, figsize=(15, 4))

df.boxplot(column='overlap_ratio', by='stance', ax=axes[0])
axes[0].set_title('Lexical Overlap Ratio')

df.boxplot(column='headline_coverage', by='stance', ax=axes[1])
axes[1].set_title('Headline Coverage')

df.boxplot(column='body_coverage', by='stance', ax=axes[2])
axes[2].set_title('Body Coverage')

plt.suptitle('Lexical Overlap by stance')
plt.tight_layout()
plt.show()
```



From lexical overlap, we can see that the "unrelated" category has a much lower overlap compared to the other categories. This makes sense, as unrelated articles are less likely to share common words with the headline. The "agree", "disagree", and "discuss" categories have higher overlaps, which indicates that they are more closely related to the headline.

Another thing to note is that the "discuss" and "agree" categories have a slightly higher overlap compared to the "disagree" category. This could be because articles that agree or discuss the headline are more likely to use similar language, while articles that disagree may use different language to express their opposing views.

1.1.5 Data quality and missing values

[69]:
```
# Missing values
print("Missing values:")
print(df.isnull().sum())

# Duplicates
print(f"\nTotal duplicate rows: {df.duplicated().sum()}")
print(f"Duplicate headline-body pairs: {df.duplicated(subset=['headline',␣
↪'body']).sum()}")

# Extremely short texts
print(f"\nHeadlines < 5 words: {(df['headline_length_words'] < 5).sum()}")
print(f"Bodies < 10 words: {(df['body_length_words'] < 10).sum()}")
print(f"Empty headlines: {df['headline'].str.strip().eq('').sum()}")
print(f"Empty bodies: {df['body'].str.strip().eq('').sum()}")
```

```
Missing values:
headline                 0
stance                   0
body                     0
headline_length_words    0
headline_length_chars    0
body_length_words        0
body_length_chars        0
overlap_ratio            0
headline_coverage        0
body_coverage            0
dtype: int64

Total duplicate rows: 713
Duplicate headline-body pairs: 713

Headlines < 5 words: 994
Bodies < 10 words: 248
Empty headlines: 0
Empty bodies: 0
```

So, we have some duplicates in the dataset. We also have headlines and bodies that are very short. No empty ones though.

[70]:
```
# show duplicate headline-body pairs
duplicate_pairs = df[df.duplicated(subset=['headline', 'body'], keep=False)]
duplicate_pairs.head()
```

[70]:
```
                                              headline     stance  \
48                Rumors of ISIS leader's death debunked  unrelated
193  Michelle Obama’s face blurred by Saudi state t…  unrelated
```

```
229  Meteorite leaves crater in Nicaraguan capital …  unrelated
340      Saudi cleric condemns snowmen as anti-Islamic  unrelated
369  Rumor: Gold Apple Watch Edition priced up to $…    discuss

                                                  body  headline_length_words  \
48   The CBI registered a case against Gurmeet Ram …                       6
193  Over the weekend there was a rumor flying arou…                       8
229  CHILPANCINGO, Mexico (AP) - Authorities testin…                      7
340  Mounted with a cannon and leaving a trail of s…                      6
369  When Apple introduced its Apple Watch in Septe…                      18

     headline_length_chars  body_length_words  body_length_chars  \
48                      38                255               1572
193                     55                213               1163
229                     53               1433               9189
340                     45                559               3293
369                     97                336               1972

     overlap_ratio  headline_coverage  body_coverage
48        0.006944           0.166667       0.007194
193       0.006757           0.125000       0.007092
229       0.003929           0.285714       0.003968
340       0.000000           0.000000       0.000000
369       0.043062           0.500000       0.045000
```

[71]:

```python
# group and count duplicate groups
dup_groups = duplicate_pairs.groupby(['headline','body']).size().
 ↪reset_index(name='count').sort_values('count', ascending=False)
dup_groups
```

[71]:

```
                                              headline  \
89   Disturbed aunt cuts off nephew’s penis after h…
534  Suspended boy threatened to make student disap…
560  Two GOP Congressmen Say Suspected Terrorists C…
103                        Evil Aunt Hacks Off Toddlers Penis
474          Source: Joan Rivers' doc did biopsy, selfie
..                                                 …
239  Joan Rivers Personal Doctor Allegedly Took A S…
240  Joan Rivers Personal Doctor Allegedly Took A S…
241  Joan Rivers Personal Doctor Allegedly Took A S…
242  Joan Rivers Personal Doctor Allegedly Took A S…
705  ‘I will have justice for what they did to me’:…

                                                  body  count
89   A wicked aunt cut her three-year-old nephew’s …      4
534  A Grade 4 student was reportedly suspended fro…      3
560  A Texas National Guard soldier scans the Mexic…      3
```

```
103   A wicked aunt cut her three-year-old nephew's …      3
474   (CNN) -The cardiac arrest leading to Joan Rive…      3
..                                                …    …
239   Raven-Symone did NOT file molestation charges …      2
240   Rivers, 81, had checked in for routine throat …      2
241   TEHRAN (FNA)- Iraq's army has shot down two Br…      2
242   THE scourge of ISIS - a female Kurdish warrior…      2
705   A Facebook post By Tikal goldie showed a image…      2

[706 rows x 3 columns]
```

All these duplicates are the result of headline-body mix to create more unrelated samples.

[72]:
```
# Healines under 5 words and bodies under 10 words are mostly duplicates.

df[(df['headline_length_words'] < 5) & (df['body_length_words'] < 10)]
```

[72]:
```
                              headline      stance  \
6180                ISIS leader dead?   unrelated
21578  Batmobile wasn't stolen: Cops   unrelated
43016               ISIS leader dead?     discuss
57809                  Breast Chancer     discuss
59311           President Sisi's Gift   unrelated
59318           3-Boobed Woman a Fake       agree
63053         Meet the 3-boobed woman    disagree

                                              body  \
6180             I aborted my baby because it was a boy
21578            I aborted my baby because it was a boy
43016  A US airstrike allegedly killed ISIS leader Al…
57809                 The third breast story is a hoax.
59311                    Microsoft will buy Mojang AB.
59318                 The third breast story is a hoax.
63053                 The third breast story is a hoax.

       headline_length_words  headline_length_chars  body_length_words  \
6180                       3                     17                  9
21578                      4                     29                  9
43016                      3                     17                  8
57809                      2                     14                  7
59311                      3                     21                  5
59318                      4                     21                  7
63053                      4                     23                  7

       body_length_chars  overlap_ratio  headline_coverage  body_coverage
6180                  38       0.000000           0.000000       0.000000
21578                 38       0.000000           0.000000       0.000000
43016                 56       0.222222           0.666667       0.250000
```

```
57809                      33       0.125000          0.500000       0.142857
59311                      29       0.000000          0.000000       0.000000
59318                      33       0.100000          0.250000       0.142857
63053                      33       0.100000          0.250000       0.142857
```

Some of these make sense, I think maybe we can keep these rows.

[73]:
```python
# Headlines under 5 words

df[df['headline_length_words'] < 5]
```

[73]:
```
                             headline     stance  \
10                     Gateway Pundit    discuss
42                  Seth Rogen Is Woz  unrelated
57                          Bali Awry  unrelated
73     Batmobile wasn't stolen: Cops  unrelated
237                 ISIS Getting Ebola  unrelated
…                                 …        …
75374            Why Obamacare failed   disagree
75375            Why Obamacare failed    discuss
75376            Why Obamacare failed      agree
75377            Why Obamacare failed      agree
75378            Why Obamacare failed   disagree

                                                    body  \
10     A British rapper whose father is awaiting tria…
42     Last year, a Vine from President Obama's trip …
57     While Apple announced that the base model of i…
73     The news has gone around the world, including …
237    DNA tests have confirmed that a daughter and a…
…                                                    …
75374  Congressional Republicans, evidently hoping th…
75375  Did Obamacare work?\r\n\r\nIt's worth reflecti…
75376  Millions may lose coverage next year if Congre…
75377  Come November, the grim trudge across the incr…
75378  Remember how much Republicans wanted to repeal…

       headline_length_words  headline_length_chars  body_length_words  \
10                         2                     14                458
42                         4                     17                 61
57                         2                      9                195
73                         4                     29                377
237                        3                     18                516
…                        …                     …                 …
75374                      3                     20               1055
75375                      3                     20               1658
75376                      3                     20               1001
75377                      3                     20                909
```

```
75378                          3                   20                 813

       body_length_chars  overlap_ratio  headline_coverage  body_coverage
10                  2752       0.000000           0.000000       0.000000
42                   396       0.017241           0.250000       0.018182
57                  1113       0.000000           0.000000       0.000000
73                  2115       0.000000           0.000000       0.000000
237                 3142       0.000000           0.000000       0.000000
…                      …          …                 …              …
75374               6525       0.001923           0.333333       0.001931
75375              10376       0.002853           0.666667       0.002857
75376               6197       0.002222           0.333333       0.002232
75377               5720       0.006110           1.000000       0.006110
75378               5043       0.002155           0.333333       0.002165

[994 rows x 10 columns]
```

[74]:
```
# single word headlines
df[df['headline_length_words'] == 1]
```

[74]:
```
        headline     stance  \
50110  Crabzilla  unrelated
50409  Crabzilla  unrelated
51000  Crabzilla  unrelated
51179  Crabzilla  unrelated
51241  Crabzilla  unrelated
52218  Crabzilla  unrelated
52730  Crabzilla    discuss
53038  Crabzilla    discuss
53632  Crabzilla    discuss
53943  Crabzilla    discuss
55584  Crabzilla  unrelated
55774  Crabzilla  unrelated
56622  Crabzilla  unrelated
56641  Crabzilla  unrelated
58631  Crabzilla  unrelated
59194  Crabzilla    discuss
59421  Crabzilla  unrelated
59537  Crabzilla    discuss
60054  Crabzilla    discuss
60377  Crabzilla  unrelated
60393  Crabzilla    discuss
60860  Crabzilla  unrelated
61271  Crabzilla    discuss
61983  Crabzilla  unrelated
63999  Crabzilla  unrelated
64521  Crabzilla  unrelated
```

```
65090  Crabzilla    discuss
65654  Crabzilla    discuss
65894  Crabzilla  unrelated
65997  Crabzilla    discuss
66152  Crabzilla  unrelated
66567  Crabzilla    discuss
68630  Crabzilla  unrelated
69829  Crabzilla  unrelated
70248  Crabzilla  unrelated
70430  Crabzilla  unrelated
71668  Crabzilla    discuss
72077  Crabzilla  unrelated
72600  Crabzilla    discuss
73156  Crabzilla    discuss
74238  Crabzilla  unrelated
74418  Crabzilla  unrelated
75003  Crabzilla  unrelated

                                                    body  \
50110  Police are looking for the married lovers of s…
50409  British jihadist al-Britani was reportedly kil…
51000  The Apple Watch will reportedly be available i…
51179  Either Lenovo is very serious about stepping u…
51241  Apple originally planned for the Apple Watch t…
52218  Tonight's Australian Open coverage on ESPN2 fe…
52730  THIS seaside town might be famed for its oyste…
53038  A marine biologist has killed off claims that …
53632  The crabs human see or eat are usually only si…
53943  "The aerial shot enables viewers to see the fu…
55584  Seems like the President of Argentina Cristina…
55774  When Tim Cook hits the stage Monday to reveal …
56622  Police are hunting four masked men who burst i…
56641  Amid conflicting reports, an Islamic State (Is…
58631  The President of Argentina, Cristina Fernandez…
59194  Is it a bird? Is it a plane? Or is it a giant …
59421  The hackers behind a devastating cyberattack a…
59537  AUSTIN (KXAN) - This giant news is a colossal …
60054  An astonishing image appears to show a giant c…
60377  With Apple looking to take some serious securi…
60393  Nowadays, it seems the world is just littered …
60860  Once thought to be a key component of Apple's …
61271  The seaside town might be famed for its oyster…
61983  As traditions go this may be the most unbeliev…
63999  A five-year-old has been handed an invoice and…
64521  nce, War\r\nThe Islamic State is harvesting hu…
65090  The man behind the world famous 50ft Crabzilla…
65654  Claim: An aerial photograph shows a giant 50-f…
```

```
65894    Brian Stelter from CNN just reported that hack…
65997    IS it Crabzilla or Claws? This aerial image of…
66152    This is the stuff nightmares are made of, but …
66567    Leading invertebrate researcher says "a hoax i…
68630    SEOUL, Feb. 5 (Korea Bizwire) - A powerful rob…
69829    While protesters in Hong Kong seem to have ado…
70248    A clinic employee tells investigators the pict…
70430    (CNN) -- Joan Rivers' personal throat doctor d…
71668    Debate is raging online as to whether pic show…
72077    Although Joan Rivers went to Yorkville Endosco…
72600    A giant crab photographed in a harbour mouth i…
73156    Does crabzilla really exist? A Mysterious 50ft…
74238    Details surrounding the death of comedian Joan…
74418    Apple is the sort of company to be so obsessiv…
75003    "How nice pants" (referring to the rear) is pr…

       headline_length_words  headline_length_chars  body_length_words  \
50110                      1                      9                158
50409                      1                      9                 13
51000                      1                      9                225
51179                      1                      9                416
51241                      1                      9                289
52218                      1                      9                 98
52730                      1                      9                283
53038                      1                      9                457
53632                      1                      9                199
53943                      1                      9                114
55584                      1                      9                305
55774                      1                      9                765
56622                      1                      9                226
56641                      1                      9                289
58631                      1                      9                211
59194                      1                      9                148
59421                      1                      9                340
59537                      1                      9                160
60054                      1                      9                553
60377                      1                      9                426
60393                      1                      9                261
60860                      1                      9                407
61271                      1                      9                364
61983                      1                      9                271
63999                      1                      9                567
64521                      1                      9                435
65090                      1                      9                419
65654                      1                      9                504
65894                      1                      9                 73
65997                      1                      9                249
```

66152	1	9	285
66567	1	9	406
68630	1	9	220
69829	1	9	24
70248	1	9	282
70430	1	9	1104
71668	1	9	252
72077	1	9	201
72600	1	9	181
73156	1	9	46
74238	1	9	236
74418	1	9	430
75003	1	9	333

	body_length_chars	overlap_ratio	headline_coverage	body_coverage
50110	843	0.000000	0.0	0.000000
50409	78	0.000000	0.0	0.000000
51000	1300	0.000000	0.0	0.000000
51179	2393	0.000000	0.0	0.000000
51241	1731	0.000000	0.0	0.000000
52218	607	0.000000	0.0	0.000000
52730	1621	0.000000	0.0	0.000000
53038	2575	0.004202	1.0	0.004202
53632	1078	0.000000	0.0	0.000000
53943	647	0.000000	0.0	0.000000
55584	1835	0.000000	0.0	0.000000
55774	4503	0.000000	0.0	0.000000
56622	1214	0.000000	0.0	0.000000
56641	1822	0.000000	0.0	0.000000
58631	1283	0.000000	0.0	0.000000
59194	836	0.000000	0.0	0.000000
59421	2107	0.000000	0.0	0.000000
59537	902	0.000000	0.0	0.000000
60054	3325	0.003378	1.0	0.003378
60377	2533	0.000000	0.0	0.000000
60393	1529	0.005525	1.0	0.005525
60860	2527	0.000000	0.0	0.000000
61271	2067	0.000000	0.0	0.000000
61983	1613	0.000000	0.0	0.000000
63999	3094	0.000000	0.0	0.000000
64521	2874	0.000000	0.0	0.000000
65090	2509	0.004016	1.0	0.004016
65654	3055	0.003425	1.0	0.003425
65894	428	0.000000	0.0	0.000000
65997	1426	0.006098	1.0	0.006098
66152	1618	0.000000	0.0	0.000000
66567	2395	0.000000	0.0	0.000000

```
68630                 1340     0.000000                0.0        0.000000
69829                  148     0.000000                0.0        0.000000
70248                 1819     0.000000                0.0        0.000000
70430                 7073     0.000000                0.0        0.000000
71668                 1460     0.000000                0.0        0.000000
72077                 1286     0.000000                0.0        0.000000
72600                 1036     0.000000                0.0        0.000000
73156                  272     0.025000                1.0        0.025000
74238                 1459     0.000000                0.0        0.000000
74418                 2506     0.000000                0.0        0.000000
75003                 2079     0.000000                0.0        0.000000
```

`Carbzilla`: it's the only one-word headline, and it is a real word, so I think we can keep it. There are some discusses there

[75]:
```python
# 10 words bodies

df[df['body_length_words'] <= 10]
```

[75]:
```
                                                headline     stance  \
383    Report: Former Baseball Star Jose Canseco Acci…  unrelated
623    Zack Snyder Makes Fun of Stolen Batmobile Rumo…  unrelated
1236   Cameron vows to hunt down IS 'monsters' after …    discuss
1370   Zack Snyder's 'Batman V Superman: Dawn Of Just…  unrelated
1403   Kidnapped Nigerian schoolgirls: Government cla…  unrelated
…                                                    …          …
74656  Next-gen iPhone rumored to sport dual-lenses, …  unrelated
74778  Fox Affiliate Falls for College Humor's Office…      agree
74979  The Story of the 3-Boobed Lady Is As Fake As H…      agree
75062  Charles Manson's fiancee used him for his … …  unrelated
75086  Woman Adds Third Breast by Undergoing Surgery …   disagree

                                                    body  \
383    Google has bought about half of Pacific Shores…
623               I aborted my baby because it was a boy
1236   An unverified video shows the beheading of Dav…
1370              I aborted my baby because it was a boy
1403   Google has bought about half of Pacific Shores…
…                                                    …
74656  Egyptian President al-Sisi reportedly offered …
74778  This Is The Most Passive-Aggressive Office Bat…
74979                The third breast story is a hoax.
75062                The third breast story is a hoax.
75086                The third breast story is a hoax.

       headline_length_words  headline_length_chars  body_length_words  \
383                       10                     67                 10
623                       11                     62                  9
```

```
1236                  10                59                 9
1370                  14                85                 9
1403                  16               115                10
…                  …                …                 …
74656                 13                88                10
74778                 10                61                10
74979                 13                59                 7
75062                  9                53                 7
75086                 12                76                 7

       body_length_chars  overlap_ratio  headline_coverage  body_coverage
383                   59       0.000000           0.000000       0.000000
623                   38       0.000000           0.000000       0.000000
1236                  55       0.055556           0.100000       0.111111
1370                  38       0.000000           0.000000       0.000000
1403                  59       0.000000           0.000000       0.000000
…                  …             …                  …              …
74656                 71       0.045455           0.076923       0.100000
74778                 65       0.052632           0.100000       0.100000
74979                 33       0.285714           0.363636       0.571429
75062                 33       0.000000           0.000000       0.000000
75086                 33       0.117647           0.166667       0.285714

[422 rows x 10 columns]
```

[76]:

```python
df[df['body_length_words'] <= 10].groupby(['headline','body']).size().
↪reset_index(name='count').sort_values('count', ascending=False)
```

[76]:

```
                                              headline  \
318   Suspended boy threatened to make student disap…
385      Wife chops off cheating husband's penis, twice
372   WHO says reports of suspected Ebola cases in I…
0     #HairGate: iPhone 6 Customers Are Complaining …
275   Report: ISIS Leader Abu Bakr Al-Baghdadi Assas…
..                                                 …
136   HBO and Apple in Talks for $15/Month Apple TV …
135   Government fires employee who skipped work for…
134   Google to buy big chunk of Pacific Shores, ico…
133         Google to Buy Redwood City Offices for $585M
418               'Three-boobed' woman: They're not fake

                                                  body  count
318                     The third breast story is a hoax.      2
385   Hearing Unconfirmed Reports of Saudi Pipeline …      2
372   Google has bought about half of Pacific Shores…      2
0     Google has bought about half of Pacific Shores…      1
275   A US airstrike allegedly killed ISIS leader Al…      1
```

```
..                                                        …     …
136  Google has bought about half of Pacific Shores…        1
135  Google has bought about half of Pacific Shores…        1
134  Google has bought about half of Pacific Shores…        1
133  Google has bought about half of Pacific Shores…        1
418                 The third breast story is a hoax.        1

[419 rows x 3 columns]
```

Most of these bodies have different headlines, so I think we can keep them as well. They are mostly unrelated, so they won't have much impact on the model performance.

[77]:
```python
# longer bodies
df[df['body_length_words'] > 4000].groupby(['headline','body']).size().
 ↪reset_index(name='count').sort_values('count', ascending=False)
```

[77]:
```
                                            headline  \
0  12in Retina MacBook Air release date rumours: …
1             Apple 'working on 12-inch MacBook Air'
2  Apple 12-Inch MacBook Air Details Emerge; Devi…
3  Apple may launch 12-inch MacBook Air with Reti…
4  Apple's 12-Inch Retina MacBook Air Shown in Ar…
5  Is Apple about to launch a totally redesigned,…
6  Report: A Radically Redesigned 12-Inch MacBook…
7  Report: new 12-inch MacBook Air slims down, ha…

                                                body  count
0  Our Retina MacBook Air rumour article brings t…      1
1  Our Retina MacBook Air rumour article brings t…      1
2  Our Retina MacBook Air rumour article brings t…      1
3  Our Retina MacBook Air rumour article brings t…      1
4  Our Retina MacBook Air rumour article brings t…      1
5  Our Retina MacBook Air rumour article brings t…      1
6  Our Retina MacBook Air rumour article brings t…      1
7  Our Retina MacBook Air rumour article brings t…      1
```

These are nothing special, just some long bodies, we can keep them.

[78]:
```python
# non-english headline or body
```

1.1.6 Cleaning and preprocessing

[79]:
```python
import re
import pandas as pd
import nltk
from nltk.corpus import stopwords
from nltk.stem import WordNetLemmatizer
from nltk import word_tokenize, pos_tag
```

```python
from nltk.corpus import wordnet

nltk.download('punkt')
nltk.download('stopwords')
nltk.download('wordnet')
nltk.download('averaged_perceptron_tagger')
nltk.download('omw-1.4')
nltk.download('averaged_perceptron_tagger')

stop_words = set(stopwords.words('english'))
lemmatizer = WordNetLemmatizer()
```

```
[nltk_data] Downloading package punkt to
[nltk_data]     C:\Users\HP\AppData\Roaming\nltk_data…
[nltk_data]   Package punkt is already up-to-date!
[nltk_data] Downloading package stopwords to
[nltk_data]     C:\Users\HP\AppData\Roaming\nltk_data…
[nltk_data]   Package stopwords is already up-to-date!
[nltk_data] Downloading package wordnet to
[nltk_data]     C:\Users\HP\AppData\Roaming\nltk_data…
[nltk_data]   Package wordnet is already up-to-date!
[nltk_data] Downloading package averaged_perceptron_tagger to
[nltk_data]     C:\Users\HP\AppData\Roaming\nltk_data…
[nltk_data]   Package averaged_perceptron_tagger is already up-to-
[nltk_data]       date!
[nltk_data] Downloading package omw-1.4 to
[nltk_data]     C:\Users\HP\AppData\Roaming\nltk_data…
[nltk_data]   Package omw-1.4 is already up-to-date!
[nltk_data] Downloading package averaged_perceptron_tagger to
[nltk_data]     C:\Users\HP\AppData\Roaming\nltk_data…
[nltk_data]   Package averaged_perceptron_tagger is already up-to-
[nltk_data]       date!
```

[80]:
```python
def _pos_tag_to_wordnet(tag):
    if tag.startswith('J'):
        return wordnet.ADJ
    if tag.startswith('V'):
        return wordnet.VERB
    if tag.startswith('N'):
        return wordnet.NOUN
    if tag.startswith('R'):
        return wordnet.ADV

    return wordnet.NOUN
```

[81]:
```python
def safe_pos_tag(tokens):
    try:
        return pos_tag(tokens)
```

```
    except LookupError:
        return [(t, 'NN') for t in tokens]

def clean_text(s, remove_stopwords=True, lemmatize=True):
    s = '' if pd.isna(s) else str(s)
    s = s.lower()
    s = re.sub(r'https?://\S+|www\.\S+', ' ', s)             # URLs
    s = re.sub(r'<[^>]+>', ' ', s)                          # HTML tags
    s = re.sub(r'\S+@\S+', ' ', s)                          # emails
    s = re.sub(r'[\r\n]+', ' ', s)
    s = re.sub(r'[^0-9a-z\s]', ' ', s)                      # keep only alnum +␣
 ↪space
    s = re.sub(r'\s+', ' ', s).strip()
    if s == '':
        return ''
    tokens = word_tokenize(s)
    if remove_stopwords:
        tokens = [t for t in tokens if t not in stop_words]
    if lemmatize:
        # pos_tags = pos_tag(tokens)
        pos_tags = safe_pos_tag(tokens)
        tokens = [lemmatizer.lemmatize(t, _pos_tag_to_wordnet(pt)) for t, pt in␣
 ↪pos_tags]
    return ' '.join(tokens)
```

Adding new cleaned features

[82]:
```
df['headline_clean'] = df['headline'].map(lambda x: clean_text(x))
df['body_clean']     = df['body'].map(lambda x: clean_text(x))
df['headline_tokens'] = df['headline_clean'].str.split().map(lambda x: set(x)␣
 ↪if isinstance(x, list) else set())
df['body_tokens']     = df['body_clean'].str.split().map(lambda x: set(x) if␣
 ↪isinstance(x, list) else set())
```

[83]:
```
def overlap_from_sets(hset, bset):
    union = hset.union(bset)
    inter = hset.intersection(bset)
    if not union:
        return 0.0, 0.0, 0.0
    return len(inter)/len(union), (len(inter)/len(hset) if hset else 0.0),␣
 ↪(len(inter)/len(bset) if bset else 0.0)

df[['overlap_clean', 'headline_coverage_clean', 'body_coverage_clean']] = df.
 ↪apply(
    lambda r: pd.Series(overlap_from_sets(r['headline_tokens'],␣
 ↪r['body_tokens'])), axis=1
)
```

[84]:
```python
print(df.groupby('stance')[['overlap_ratio','overlap_clean']].
 ↪agg(['mean','median']).round(4))

# Save cleaned dataset
# df.to_csv('../data/full_clean.csv', index=False)
```

```
          overlap_ratio         overlap_clean
                   mean  median          mean  median
stance
agree            0.0444  0.0345        0.0563  0.0429
disagree         0.0364  0.0275        0.0464  0.0314
discuss          0.0406  0.0308        0.0550  0.0400
unrelated        0.0117  0.0091        0.0042  0.0000
```

[85]:
```python
df.head()
```

[85]:
```
                                            headline     stance  \
0  Police find mass graves with at least '15 bodi…  unrelated
1  Hundreds of Palestinians flee floods in Gaza a…      agree
2  Christian Bale passes on role of Steve Jobs, a…  unrelated
3  HBO and Apple in Talks for $15/Month Apple TV …  unrelated
4  Spider burrowed through tourist's stomach and …   disagree

                                                body  headline_length_words  \
0  Danny Boyle is directing the untitled film\r\n…                     19
1  Hundreds of Palestinians were evacuated from t…                     11
2  30-year-old Moscow resident was hospitalized w…                     16
3  (Reuters) - A Canadian soldier was shot at the…                     14
4  Fear not arachnophobes, the story of Bunbury's…                     10

   headline_length_chars  body_length_words  body_length_chars  overlap_ratio  \
0                    115                195               1105       0.013605
1                     65                429               2608       0.043478
2                     91                194               1123       0.028777
3                     82                 80                504       0.027027
4                     63                612               3433       0.021605

   headline_coverage  body_coverage  \
0           0.111111       0.015267
1           0.909091       0.043668
2           0.250000       0.031496
3           0.166667       0.031250
4           0.700000       0.021807

                                      headline_clean  \
0  police find mass graf least 15 body near mexic…
1  hundred palestinian flee flood gaza israel ope…
2  christian bale pass role steve job actor repor…
```

```
3  hbo apple talk 15 month apple tv streaming ser…
4                spider burrowed tourist stomach chest

                                         body_clean  \
0  danny boyle directing untitled film seth rogen…
1  hundred palestinian evacuated home sunday morn…
2  30 year old moscow resident hospitalized wound…
3  reuters canadian soldier shot canadian war mem…
4  fear arachnophobes story bunbury spiderman mig…

                                    headline_tokens  \
0  {15, find, least, student, mass, body, town, n…
1  {dam, gaza, israel, hundred, open, flee, flood…
2  {reportedly, job, part, actor, right, christia…
3  {15, tv, streaming, service, launching, apple,…
4        {stomach, chest, spider, burrowed, tourist}

                                        body_tokens  overlap_clean  \
0  {sony, anticipated, female, worked, meeting, c…       0.000000
1  {unrwa, construction, wadi, homeless, assault,…       0.040462
2  {wife, nature, told, gently, held, sad, became…       0.000000
3  {killing, reporting, amran, editing, hopkins, …       0.000000
4  {air, identification, parasitic, framenau, see…       0.014706

   headline_coverage_clean  body_coverage_clean
0                    0.000             0.000000
1                    0.875             0.040698
2                    0.000             0.000000
3                    0.000             0.000000
4                    0.600             0.014851
```

1.2 N-Gram anaysis

[86]:
```
# N-gram setup
from sklearn.feature_extraction.text import CountVectorizer

stances = ['agree', 'disagree', 'discuss', 'unrelated']
```

[87]:
```
def top_ngrams_by_stance(df, text_col, stance_col='stance', ngram_range=(1, 1),␣
↪top_k=20, min_df=2):
    rows = []
    for st in stances:
        corpus = df.loc[df[stance_col] == st, text_col].fillna('')
        vectorizer = CountVectorizer(ngram_range=ngram_range, min_df=min_df)
        X = vectorizer.fit_transform(corpus)

        terms = vectorizer.get_feature_names_out()
```

```
        counts = X.sum(axis=0).A1
        total = counts.sum()

        tmp = pd.DataFrame({
            'stance': st,
            'ngram': terms,
            'count': counts
        })
        tmp['rel_freq'] = tmp['count'] / total if total > 0 else 0
        tmp = tmp.sort_values('count', ascending=False).head(top_k)
        rows.append(tmp)

    return pd.concat(rows, ignore_index=True)
```

[88]:
```
# Headlines: top unigrams + bigrams
top_uni_head = top_ngrams_by_stance(df, text_col='headline_clean',␣
 ↪ngram_range=(1,1), top_k=20, min_df=2)
top_bi_head  = top_ngrams_by_stance(df, text_col='headline_clean',␣
 ↪ngram_range=(2,2), top_k=20, min_df=2)

display(top_uni_head.head(20))
display(top_bi_head.head(20))
```

	stance	ngram	count	rel_freq
0	agree	man	435	0.009675
1	agree	isi	424	0.009430
2	agree	woman	383	0.008518
3	agree	video	301	0.006695
4	agree	hoax	286	0.006361
5	agree	dead	264	0.005872
6	agree	foley	248	0.005516
7	agree	penis	244	0.005427
8	agree	james	232	0.005160
9	agree	president	229	0.005093
10	agree	year	221	0.004915
11	agree	say	207	0.004604
12	agree	werewolf	202	0.004493
13	agree	boy	199	0.004426
14	agree	journalist	188	0.004181
15	agree	attack	184	0.004092
16	agree	claim	181	0.004026
17	agree	death	177	0.003937
18	agree	american	175	0.003892
19	agree	bear	173	0.003848

	stance	ngram	count	rel_freq
0	agree	justin bieber	161	0.004209
1	agree	argentina president	151	0.003948

2	agree	wright foley	127	0.003321
3	agree	james wright	127	0.003321
4	agree	journalist james	124	0.003242
5	agree	year old	115	0.003007
6	agree	macaulay culkin	108	0.002824
7	agree	james foley	105	0.002745
8	agree	american journalist	105	0.002745
9	agree	islamic state	102	0.002667
10	agree	death hoax	99	0.002588
11	agree	bear attack	98	0.002562
12	agree	steve job	96	0.002510
13	agree	christian bale	94	0.002458
14	agree	bieber ringtone	88	0.002301
15	agree	beheads american	86	0.002249
16	agree	third breast	83	0.002170
17	agree	adopts jewish	81	0.002118
18	agree	spice condom	78	0.002039
19	agree	pumpkin spice	78	0.002039

[89]:
```python
# Bodies: top unigrams + bigrams
top_uni_body = top_ngrams_by_stance(df, text_col='body_clean',␣
↪ngram_range=(1,1), top_k=20, min_df=5)
top_bi_body  = top_ngrams_by_stance(df, text_col='body_clean',␣
↪ngram_range=(2,2), top_k=20, min_df=5)

display(top_uni_body.head(20))
display(top_bi_body.head(20))
```

	stance	ngram	count	rel_freq
0	agree	said	12355	0.011098
1	agree	year	5397	0.004848
2	agree	one	4813	0.004323
3	agree	would	4265	0.003831
4	agree	video	4085	0.003669
5	agree	told	3807	0.003420
6	agree	time	3666	0.003293
7	agree	people	3648	0.003277
8	agree	state	3628	0.003259
9	agree	also	3584	0.003219
10	agree	report	3514	0.003157
11	agree	news	3402	0.003056
12	agree	say	3275	0.002942
13	agree	man	3273	0.002940
14	agree	police	2872	0.002580
15	agree	new	2704	0.002429
16	agree	like	2693	0.002419
17	agree	woman	2688	0.002415
18	agree	group	2569	0.002308

```
19  agree   story   2517  0.002261
```

```
   stance                ngram  count  rel_freq
0   agree             year old   1994  0.002273
1   agree        islamic state   1631  0.001859
2   agree             new york    757  0.000863
3   agree         united state    607  0.000692
4   agree         social medium   606  0.000691
5   agree            last year    579  0.000660
6   agree       macaulay culkin   502  0.000572
7   agree           boko haram    489  0.000557
8   agree            last week    486  0.000554
9   agree          seventh son    471  0.000537
10  agree          james foley    468  0.000533
11  agree  american journalist    463  0.000528
12  agree             year ago    435  0.000496
13  agree        official said    419  0.000478
14  agree          white house    401  0.000457
15  agree         zehaf bibeau    395  0.000450
16  agree         pumpkin spice   374  0.000426
17  agree         war memorial    367  0.000418
18  agree          jihadi john    367  0.000418
19  agree             fox news    342  0.000390
```

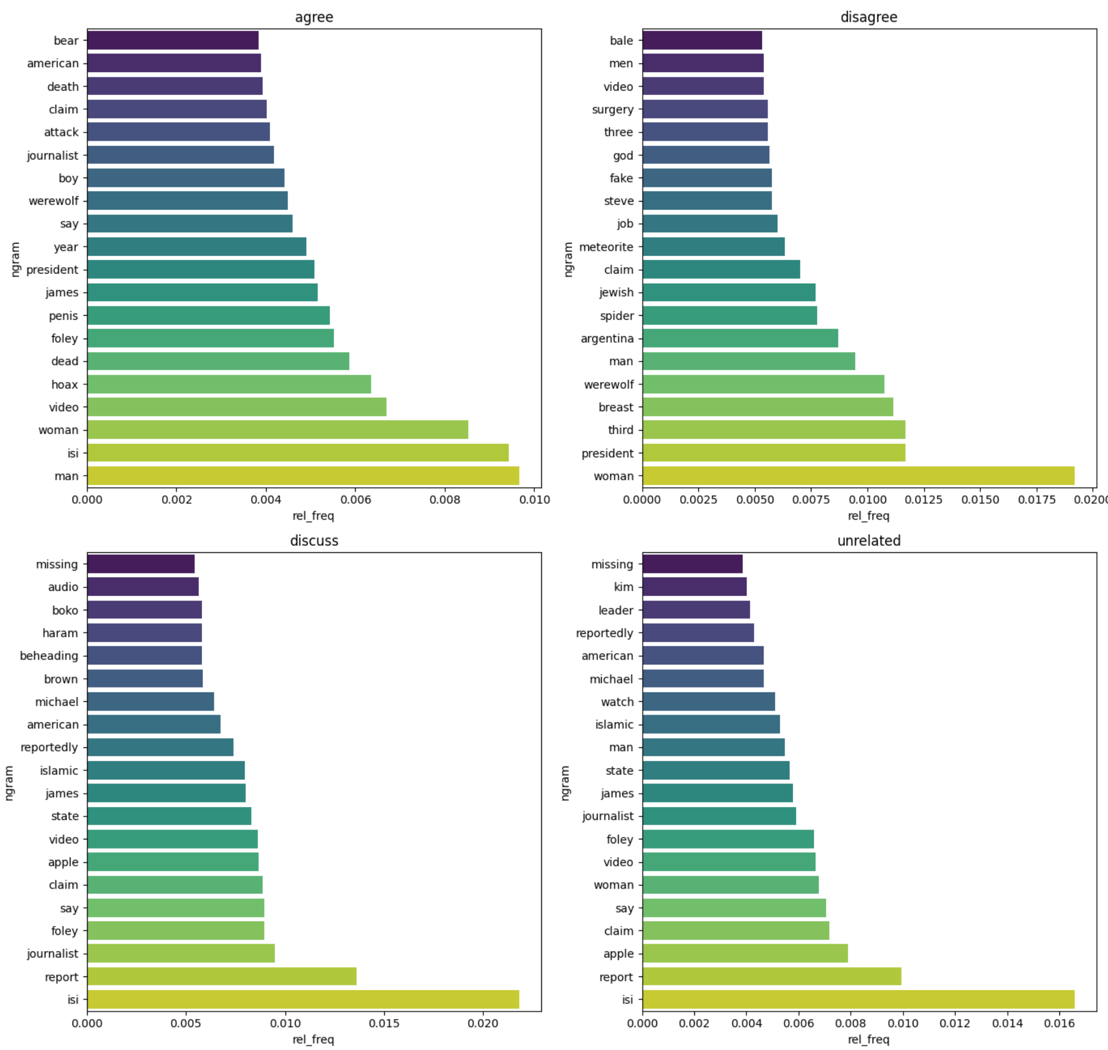
[90]:

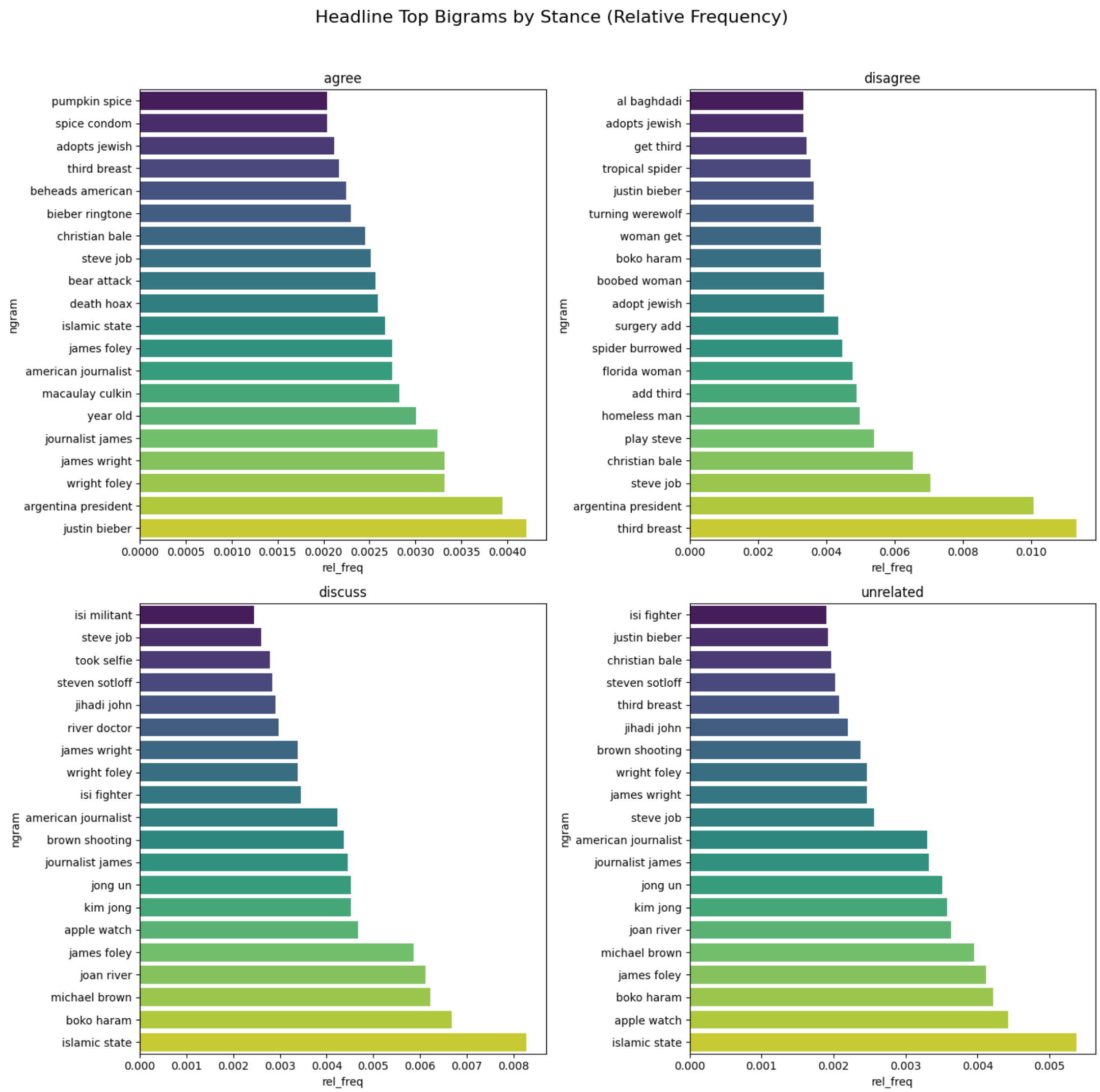
```
def plot_top_ngrams(df_top, title, metric='rel_freq'):
    import math
    stances = list(df_top['stance'].unique())
    n = len(stances)
    # number of rows when using 2 columns
    cols = 2
    rows = math.ceil(n / cols)
    # compute max bars per subplot to scale height
    max_bars = df_top.groupby('stance')['ngram'].nunique().max()
    per_plot_height = max(3.0, 0.35 * max_bars)  # adjust multiplier to taste
    figsize = (14, per_plot_height * rows)

    fig, axes = plt.subplots(rows, cols, figsize=figsize, squeeze=False)
    axes = axes.flatten()

    for i, st in enumerate(stances):
        ax = axes[i]
        sub = df_top[df_top['stance'] == st].sort_values(metric, ascending=True)
        sns.barplot(x=metric, y='ngram', data=sub, ax=ax, palette='viridis')
        ax.set_title(st)
        ax.tick_params(axis='y', labelsize=10)
        ax.set_xlabel(metric)
    # hide any unused subplots
    for j in range(len(stances), len(axes)):
```

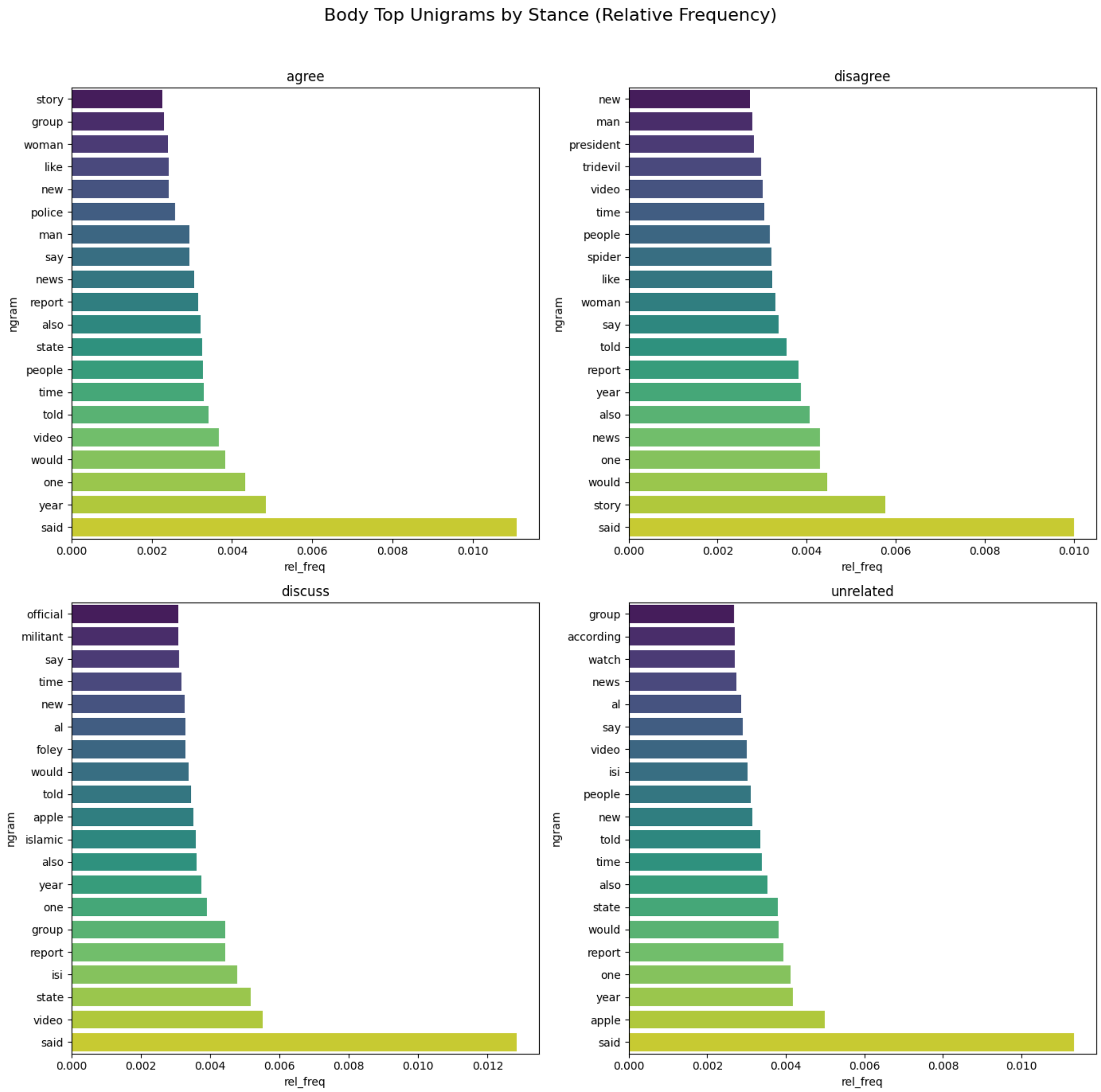
```
        axes[j].set_visible(False)

    fig.suptitle(title, fontsize=16)
    plt.tight_layout(rect=[0, 0, 1, 0.96])
    plt.show()
```

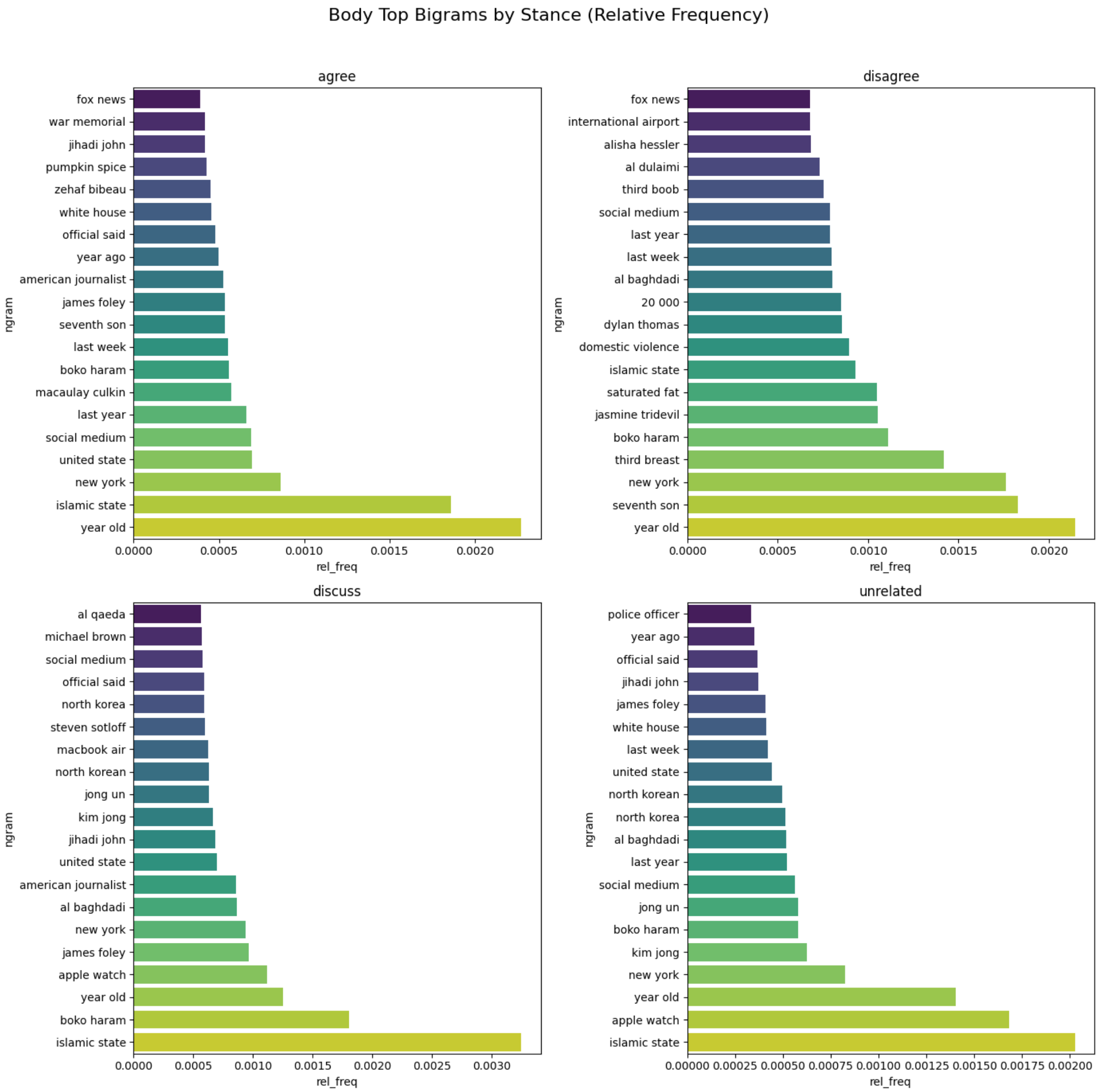
[91]:
```
plot_top_ngrams(top_uni_head, 'Headline Top Unigrams by Stance (Relative Frequency)', metric='rel_freq')
plot_top_ngrams(top_bi_head,  'Headline Top Bigrams by Stance (Relative Frequency)', metric='rel_freq')
plot_top_ngrams(top_uni_body, 'Body Top Unigrams by Stance (Relative Frequency)', metric='rel_freq')
plot_top_ngrams(top_bi_body,  'Body Top Bigrams by Stance (Relative Frequency)', metric='rel_freq')
```

Headline Top Unigrams by Stance (Relative Frequency)

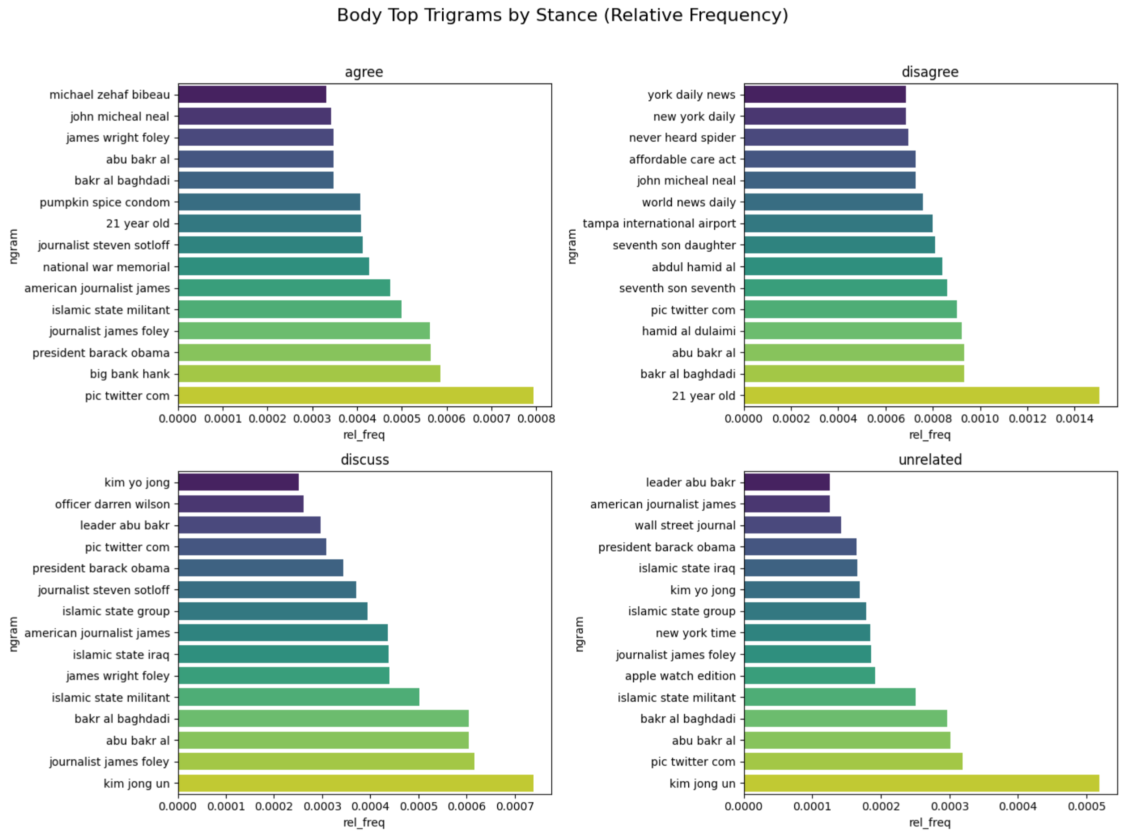



Headline Top Bigrams by Stance (Relative Frequency)
agree
pumpkin spice
spice condom
adopts jewish
third breast
beheads american
bieber ringtone
christian bale
steve job
bear attack
death hoax
islamic state
james foley
american journalist
macaulay culkin
year old
journalist james
james wright
wright foley
argentina president
justin bieber
0.0000 0.0005 0.0010 0.0015 0.0020 0.0025 0.0030 0.0035 0.0040
rel_freq
ngram
disagree
al baghdadi
adopts jewish
get third
tropical spider
justin bieber
turning werewolf
woman get
boko haram
boobed woman
adopt jewish
surgery add
spider burrowed
florida woman
add third
homeless man
play steve
christian bale
steve job
argentina president
third breast
0.000 0.002 0.004 0.006 0.008 0.010
rel_freq
ngram
discuss
isi militant
steve job
took selfie
steven sotloff
jihadi john
river doctor
james wright
wright foley
isi fighter
american journalist
brown shooting
journalist james
jong un
kim jong
apple watch
james foley
joan river
michael brown
boko haram
islamic state
0.000 0.001 0.002 0.003 0.004 0.005 0.006 0.007 0.008
rel_freq
ngram
unrelated
isi fighter
justin bieber
christian bale
steven sotloff
third breast
jihadi john
brown shooting
wright foley
james wright
steve job
american journalist
journalist james
jong un
kim jong
joan river
michael brown
james foley
boko haram
apple watch
islamic state
0.000 0.001 0.002 0.003 0.004 0.005
rel_freq
ngram


Body Top Unigrams by Stance (Relative Frequency)
agree
story
group
woman
like
new
police
man
say
news
report
also
state
people
time
told
video
would
one
year
said
ngram
rel_freq
0.000
0.002
0.004
0.006
0.008
0.010
disagree
new
man
president
tridevil
video
time
people
spider
like
woman
say
told
report
year
also
news
one
would
story
said
ngram
rel_freq
0.000
0.002
0.004
0.006
0.008
0.010
discuss
official
militant
say
time
new
al
foley
would
told
apple
islamic
also
year
one
group
report
isi
state
video
said
ngram
rel_freq
0.000
0.002
0.004
0.006
0.008
0.010
0.012
unrelated
group
according
watch
news
al
say
video
isi
people
new
told
time
also
state
would
report
one
year
apple
said
ngram
rel_freq
0.000
0.002
0.004
0.006
0.008
0.010




```
[92]: # trigrams
top_tri_body = top_ngrams_by_stance(df, text_col='body_clean',
    ngram_range=(3,3), top_k=15, min_df=10)
plot_top_ngrams(top_tri_body, 'Body Top Trigrams by Stance (Relative
    Frequency)', metric='rel_freq')
```

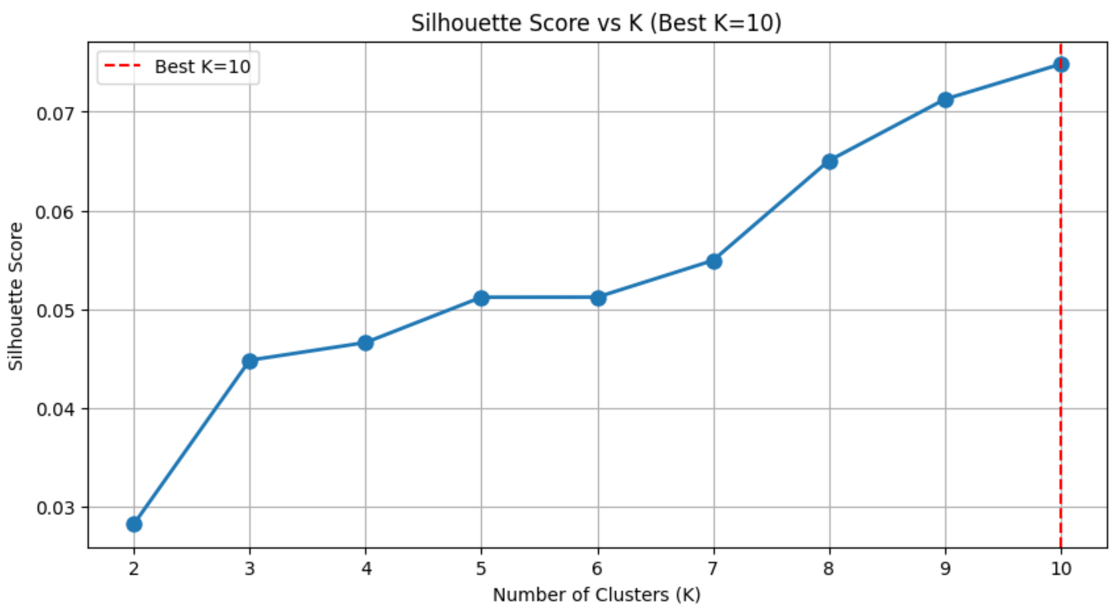


1.3 Clustering analysis

[93]:
```
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score

# Vectorize cleaned text
vectorizer = TfidfVectorizer(max_features=500, ngram_range=(1,2))
X_tfidf = vectorizer.fit_transform(df['body_clean'])

# Find optimal K using silhouette score
silhouette_scores = []
for k in range(2, 11):
    km = KMeans(n_clusters=k, random_state=42)
    labels = km.fit_predict(X_tfidf)
    score = silhouette_score(X_tfidf, labels)
    silhouette_scores.append(score)

# Fit final model
best_k = silhouette_scores.index(max(silhouette_scores)) + 2
```

```
km_final = KMeans(n_clusters=best_k, random_state=42)
df['cluster'] = km_final.fit_predict(X_tfidf)

# Visualize top words per cluster
```

[94]:
```
# Plot silhouette scores to see which K was optimal
plt.figure(figsize=(10, 5))
plt.plot(range(2, 11), silhouette_scores, marker='o', linewidth=2, markersize=8)
plt.xlabel('Number of Clusters (K)')
plt.ylabel('Silhouette Score')
plt.title(f'Silhouette Score vs K (Best K={best_k})')
plt.axvline(x=best_k, color='red', linestyle='--', label=f'Best K={best_k}')
plt.legend()
plt.grid()
plt.show()

print(f"Best number of clusters: {best_k}")
```




```
Best number of clusters: 10
```

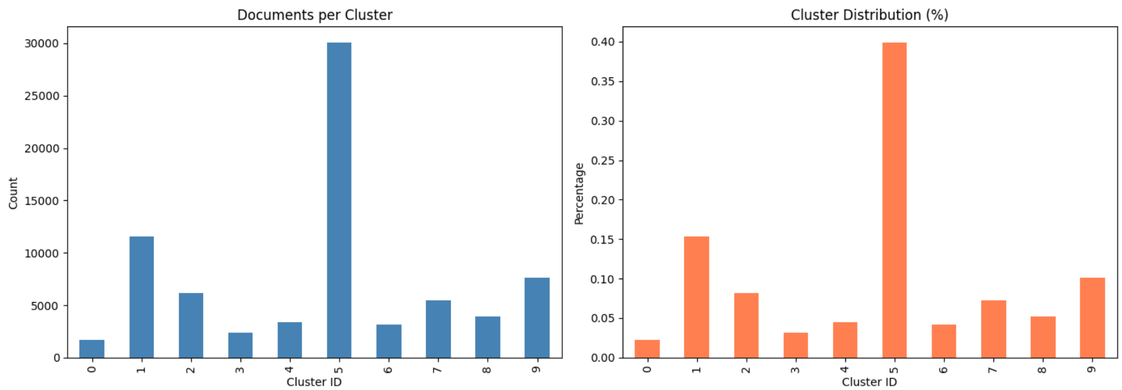
[95]:
```
def get_top_words_per_cluster(vectorizer, km_model, top_k=10):
    """Extract and display top TF-IDF words for each cluster"""
    terms = vectorizer.get_feature_names_out()
    centers = km_model.cluster_centers_

    for cluster_id in range(km_model.n_clusters):
        top_indices = centers[cluster_id].argsort()[-top_k:][::-1]
```

```
        top_words = terms[top_indices]
        print(f"\nCluster {cluster_id}: {', '.join(top_words)}")

get_top_words_per_cluster(vectorizer, km_final, top_k=15)
```

```
Cluster 0: animal, dog, said, wife, hospital, child, found, outside, woman,
year, man, find, new, old, spokesman

Cluster 1: isi, foley, islamic, video, islamic state, state, syria, journalist,
said, american, sotloff, iraq, group, border, militant

Cluster 2: apple, watch, apple watch, iphone, inch, device, company, gold, user,
model, feature, new, source, launch, store

Cluster 3: kim, jong, north, korean, korea, kim jong, jong un, un, north korean,
north korea, leader, said, official, south, health

Cluster 4: ebola, boko, boko haram, haram, girl, nigerian, government, mosul,
health, said, militant, iraqi, group, isi, official

Cluster 5: said, one, new, year, report, police, company, news, would, story,
time, people, video, job, also

Cluster 6: shot, brown, shooting, wilson, parliament, police, recording, audio,
ferguson, cnn, fired, canadian, officer, michael, said

Cluster 7: river, bear, tridevil, doctor, surgery, cat, procedure, third,
clinic, woman, said, operation, phone, told, medical

Cluster 8: al, baghdadi, al baghdadi, iraqi, strike, leader, killed, said, isi,
iraq, state, abu, islamic, group, syria

Cluster 9: boy, son, woman, girl, president, family, mother, school, parent,
child, year, said, daughter, god, old
```

[96]:
```
# Bar chart of cluster sizes
fig, axes = plt.subplots(1, 2, figsize=(14, 5))

# Absolute counts
df['cluster'].value_counts().sort_index().plot(kind='bar', ax=axes[0],␣
 ↪color='steelblue')
axes[0].set_title('Documents per Cluster')
axes[0].set_ylabel('Count')
axes[0].set_xlabel('Cluster ID')

# Percentage
```

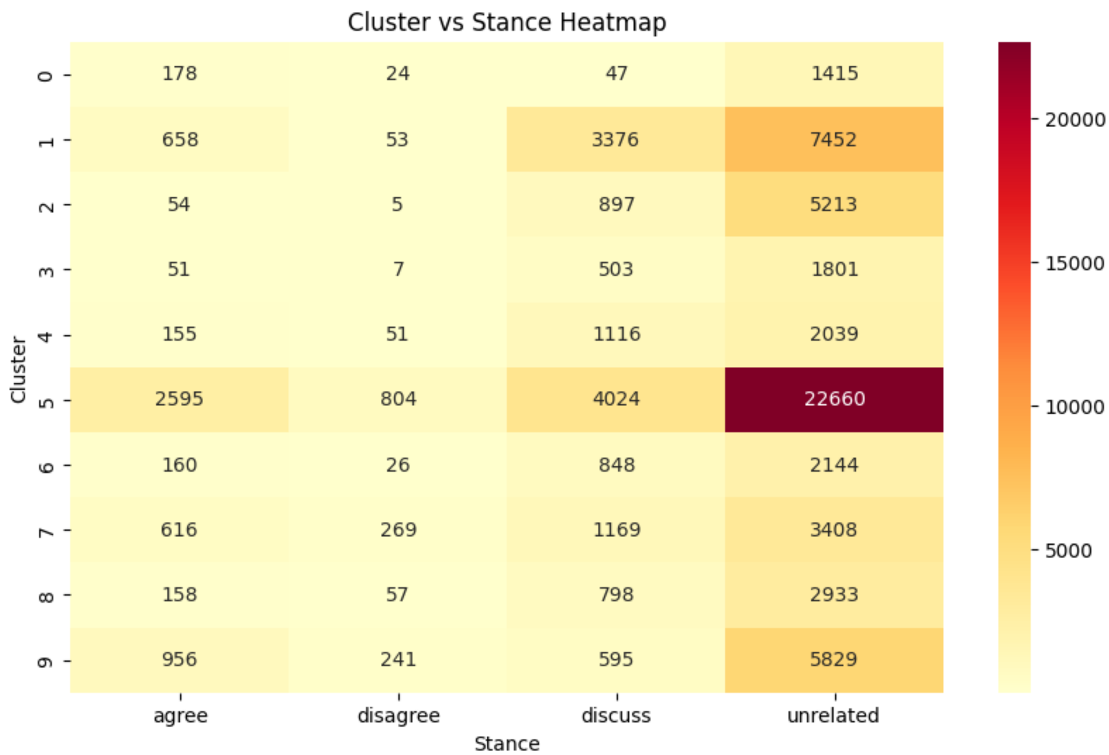
```
df['cluster'].value_counts(normalize=True).sort_index().plot(kind='bar',␣
↪ax=axes[1], color='coral')
axes[1].set_title('Cluster Distribution (%)')
axes[1].set_ylabel('Percentage')
axes[1].set_xlabel('Cluster ID')

plt.tight_layout()
plt.show()
```



[97]:
```
# See how clusters relate to stances
cluster_stance = pd.crosstab(df['cluster'], df['stance'], margins=True)
print("Cluster vs Stance Distribution:")
print(cluster_stance)

# Heatmap
plt.figure(figsize=(10, 6))
sns.heatmap(pd.crosstab(df['cluster'], df['stance']), annot=True, fmt='d',␣
↪cmap='YlOrRd')
plt.title('Cluster vs Stance Heatmap')
plt.ylabel('Cluster')
plt.xlabel('Stance')
plt.show()
```

```
Cluster vs Stance Distribution:
stance   agree  disagree  discuss  unrelated    All
cluster
0          178        24       47       1415   1664
1          658        53     3376       7452  11539
2           54         5      897       5213   6169
3           51         7      503       1801   2362
4          155        51     1116       2039   3361
5         2595       804     4024      22660  30083
6          160        26      848       2144   3178
```

7	616	269	1169	3408	5462
8	158	57	798	2933	3946
9	956	241	595	5829	7621
All	5581	1537	13373	54894	75385



[98]:
```python
# See what articles are actually in each cluster
for cluster_id in range(best_k):
    print(f"\n{'='*80}")
    print(f"CLUSTER {cluster_id} - Sample Headlines:")
    print(f"{'='*80}")
    samples = df[df['cluster'] == cluster_id][['headline', 'stance']].head(5)
    for idx, row in samples.iterrows():
        print(f"  • {row['headline'][:80]} ({row['stance']})")
```

```
================================================================================
CLUSTER 0 - Sample Headlines:
================================================================================
  • Catholic Priest Dead For 48 Minutes, Is Miraculously Revived - His
Revelations A (unrelated)
  • Pope Francis turns out not to have made pets in heaven comment (disagree)
  • Fidel Castro Dead? Yes, He Is… But The Cuban Leader Is Alive, Death Rumors
Prove (unrelated)
```

- Audio recording allegedly captures moment Michael Brown was shot (unrelated)
- Michael Phelps’ alleged girlfriend says she was born intersex (unrelated)

==

CLUSTER 1 - Sample Headlines:

==

- Accused Boston Marathon Bomber Severely Injured In Prison, May Never Walk Or Tal (unrelated)
- British Aid Worker Confirmed Murdered By ISIS (unrelated)
- Gateway Pundit (discuss)
- ISIL Beheads American Photojournalist in Iraq (discuss)
- Christian Bale Exits Steve Jobs Movie (Exclusive) (unrelated)

==

CLUSTER 2 - Sample Headlines:

==

- Identity of ISIS terrorist known as 'Jihadi John' reportedly revealed (unrelated)
- N. Korea’s Kim has leg injury but in control (unrelated)
- Bali Awry (unrelated)
- [UPDATED] 11 Men and 0 Women on Tonight's ESPN Domestic Violence Panel (unrelated)
- Paul Rudd Is Not the Viral Video Hero Who Tackled a Gay Basher in Dallas (unrelated)

==

CLUSTER 3 - Sample Headlines:

==

- TEXAS TURKEY FARM CONTAMINATED WITH EBOLA, OVER 250,000 HOLIDAY TURKEYS INFECTED (unrelated)
- SEE IT: ISIS militants caught trying to cross border into Texas, Congressman cla (unrelated)
- BREAKING NEWS: ISIS beheads missing American journalist James Wright Foley as wa (unrelated)
- Meteorite strike in Nicaragua puzzles experts (unrelated)
- 'Hairgate': iPhone 6 users encounter latest Apple problem after 'bendgate' (unrelated)

==

CLUSTER 4 - Sample Headlines:

==

- Boko Haram Denies Nigeria Cease-Fire Claim (discuss)
- Boko Haram ceasefire ignored as violence flares in Borno state (discuss)
- Insurgents killed in Nigeria despite alleged truce with Boko Haram (discuss)
- Would-be rapist has penis severed by angry mob (unrelated)
- Nigeria and Boko Haram 'agree ceasefire and girls' release' (unrelated)

==

CLUSTER 5 - Sample Headlines:
==
- Police find mass graves with at least '15 bodies' near Mexico town where 43 stud (unrelated)
- Hundreds of Palestinians flee floods in Gaza as Israel opens dams (agree)
- Spider burrowed through tourist's stomach and up into his chest (disagree)
- 'Nasa Confirms Earth Will Experience 6 Days of Total Darkness in December' Fake (agree)
- Banksy 'Arrested & Real Identity Revealed' Is The Same Hoax From Last Year (agree)

==
CLUSTER 6 - Sample Headlines:
==
- HBO and Apple in Talks for $15/Month Apple TV Streaming Service Launching in Apr (unrelated)
- Why the Gold Apple Watch Edition Must Cost $10,000 (unrelated)
- Amazon's first brick-and-mortar store said to open in Manhattan (unrelated)
- ISLAMIC STATE BEHEADS MISSING AMERICAN JOURNALIST JAMES WRIGHT FOLEY (unrelated)
- Moment Michael Brown was shot dead 'caught on audio recording': FBI handed poten (discuss)

==
CLUSTER 7 - Sample Headlines:
==
- Christian Bale passes on role of Steve Jobs, actor reportedly felt he wasn't rig (unrelated)
- Batmobile wasn't stolen: Cops (unrelated)
- Actor has his testicles stolen after meeting ‘young blonde woman at a bar’ (discuss)
- KIM DINE WITH ME: Porky president Kim Jong-un to open UK restaurant serving DOG (unrelated)
- Did Comcast Get a Man Fired From His Job for Complaining About Its Service? (unrelated)

==
CLUSTER 8 - Sample Headlines:
==
- Woman detained in Lebanon is not al-Baghdadi's wife, Iraq says (agree)
- Report: ISIS Leader Abu Bakr Al-Baghdadi Assassinated In U.S. Airstrike (discuss)
- Islamic State leader's family detained by Lebanon, officials say (discuss)
- Scientists Doubt That Meteor Caused Crater In Nicaragua (unrelated)
- Isis leader Abu Bakr al-Baghdadi's 'wife and son detained in Lebanon' (unrelated)

==

```
CLUSTER 9 - Sample Headlines:
================================================================================
  • NET Extra: Back-from-the-dead Catholic priest claims God is a female (agree)
  • Luke Somers Dies In Rescue Attempt, Sister Says; Yemeni Defense Ministry
Says Un (unrelated)
  • Nasa questions whether crater in Nicaragua caused by meteorite (unrelated)
  • Planetary Alignment On Jan 4, 2015 Will Decrease Gravity For 5 Minutes
Causing P (unrelated)
  • Michael Brown shooting audio caught on tape? (unrelated)
```

[ ]:

[ ]: